

Predicting Next-Quarter Debt Servicing Risk of Philippine Listed Companies Using Financial Statements and Financial News Sentiment Analysis

Using Supervised Machine Learning

Appendix L — Technical Deck

Business context

- Financial institutions and regulators need an early warning system capable of identifying companies that may experience debt servicing difficulties in the next quarter.
- Traditional financial ratio analysis captures historical performance but often misses qualitative market signals reflected in financial news.

Data Science Problem

Develop a machine learning model that predicts next-quarter debt servicing risk using:

- Financial statements
- Financial ratios
- Macroeconomic indicators
- Financial news sentiment (FinBERT).

Problem statement (technical)

This study aims to determine whether machine learning models can predict the next-quarter debt servicing risk of Philippine companies using quarterly financial statement data and macroeconomic indicators, and whether financial news sentiment can improve performance.

ML task type

- Supervised classification (binary for intervention)
- Target: TARGET_NEXT_QTR
 - ✓ High Risk
 - ✓ Low Risk
- Target constructed using next-quarter Debt Service Proxy Ratio (DSPR).

Success metrics

- ✓ ROC-AUC (primary)
- ✓ Recall
- ✓ Accuracy
- ✓ Precision
- ✓ F1-score
- ✓ Brier Score

Primary stakeholders

-  Banks and Lending Institutions
-  Bangko Sentral ng Pilipinas (BSP) and Financial Regulators
-  Credit Risk Analysts
-  Investment Institutions
-  Corporate Treasury and Risk

Secondary Stakeholders

- Philippine listed companies
- Credit rating agencies
- Academic researchers AI and data science practitioners
- Financial technology (FinTech) companies

ML Task

- ✓ Binary Classification

Key Benefit

- Provide an AI-powered early warning system for next-quarter debt servicing risk to support more informed lending, supervisory, and investment decisions.

Structured Dataset

- Bloomberg financial data
- Quarterly financial statements
- Macroeconomic indicators

Unstructured Dataset

Collected from

- BusinessWorld
- Philippine Daily Inquirer
- Manila Bulletin
- ABS-CBN News
- GMA News
- PSE EDGE
- SEC
- Company Investor Relations

Data Set Summary

Item	Value
Companies	30
Period	Q4 2020 – Q1 2026
News Articles	857
Company-Quarter Observations	660

Data Quality Assessment

- ✓ Missing values checked
- ✓ Duplicate detection
- ✓ Outlier analysis
- ✓ Temporal consistency
- ✓ Data type validation

The final dataset contained no missing company-quarter observations and no duplicate records.

Data dictionary — excerpt

Variable	Type
Current Assets	Numeric
Total Debt	Numeric
Revenue	Numeric
ROA	Numeric
GDP Growth	Numeric
Mean Sentiment	Numeric
News Count	Integer
TARGET_NEXT_QTR	Binary

1. DATA SOURCES

Structured:

- Financial Statements
- Macroeconomic Data
- Bloomberg
- SEC/PSE

Unstructured:

- BusinessWorld
- Inquirer
- Manila Bulletin
- ABS-CBN
- GMA
- 857 News Articles

2. DATA INTEGRATION

Financial Pipeline
+
Sentiment Pipeline
(FinBERT)

Merged Quarterly
Dataset
30 Companies
2020Q4–2026Q1
660 Company-
Quarter Observations

3. DATA PREPROCESSING

Financial:

- Missing Values
- IQR Outliers
- Scaling
- Feature Selection

Sentiment:

- Text Cleaning
- Deduplication
- Tokenization
- Confidence Validation

4. HYBRID FEATURE ENGINEERING

Financial Features:

- Liquidity
- Profitability
- Leverage
- Cash Flow
- Growth
- Macro

Sentiment Features:

- Positive
- Negative
- Neutral
- Net Sentiment
- News Volume
- Article Counts

5. FINAL MODELING DATASET

Hybrid AI Dataset

29 Financial Predictors
+ 7 Sentiment Predictors
= 36 Features

Target:

TARGET_NEXT_QTR
Bottom 25% DSPR

Time Split:

Train:2020Q4-2025Q4
Forecast:2026Q1

Data Cleaning

- Missing value handling
- Duplicate removal
- Winsorization for outliers
- Feature scaling (Logistic Regression)

Domain-derived Features

- Financial ratios
- Current Ratio
- Quick Ratio
- Debt-to-Equity
- ROA
- Revenue Growth
- Operating Cash Flow Growth

Sentiment Features

- Mean Sentiment
- Positive %
- Negative %
- News Count
- Sentiment Volatility

FinBERT outputs were aggregated into quarterly sentiment indicators and merged with the financial dataset using Company, Year, and Quarter.

Exploratory Data Analysis

Performed

- Distribution analysis
- Correlation analysis
- Boxplots
- Outlier analysis
- Feature relationships
- Class distribution

Observations

- Financial variables exhibited varying distributions.
- Outliers were expected in financial ratios
- .No duplicate observations.
- Complete dataset after preprocessing.

Relationships & correlation

- Histogram
- Correlation Heatmap
- Boxplots.

Key outcome of EDA

Quality Assessment: The dataset was complete after preprocessing, with missing values, duplicate records, and inconsistent entries successfully addressed.

Distribution Analysis: Several financial variables exhibited skewed distributions and extreme values, which are typical in corporate financial data.

Outlier Detection: Significant outliers were identified in financial ratios and cash flow variables. Winsorization was applied to reduce their impact while preserving meaningful observations.

Correlation Analysis: Moderate correlations were observed among some financial variables, but no severe multicollinearity was detected, allowing most engineered features to be retained.

Class Distribution: The target variable showed class imbalance between high-risk and low-risk companies, leading to the use of balanced learning strategies during model training.

Feature Relationships: Strong relationships were observed between debt-related ratios, profitability measures, liquidity indicators, and the target variable, supporting their inclusion in the predictive models.

Sentiment Data Integration: Quarterly news sentiment provided complementary information to financial variables, enriching the dataset with market perception and qualitative signals.

Feature Selection

- Three-stage selection
- Correlation filtering
- Embedded L1 Regularization
- Model-based importance.

Explainability

- SHAP
 - Used to explain
 - ✓ Global feature importance
 - ✓ Local predictions.

Key Findings

Top predictors

- Interest Expense
- Revenue
- ROA
- Current Assets
- GDP Growth

Note: Sentiment variables contributed additional context but were not among the strongest predictors.

Dimensionality Reduction

PCA

Purpose: Reduce Redundancy, Assess variance structure, Visualize feature space

Decision: PCA was explored for visualization and redundancy assessment but was not used in the final predictive models because preserving the interpretability of financial variables and SHAP explanations was more important for this application.

Justification: Financial risk models require interpretable variables for regulatory and business decision-making.

Supervised Models

- ✓ Logistic Regression
- ✓ Random Forest
- ✓ XGBoost

Reason: The research objective required predicting predefined High-Risk and Low-Risk classes.

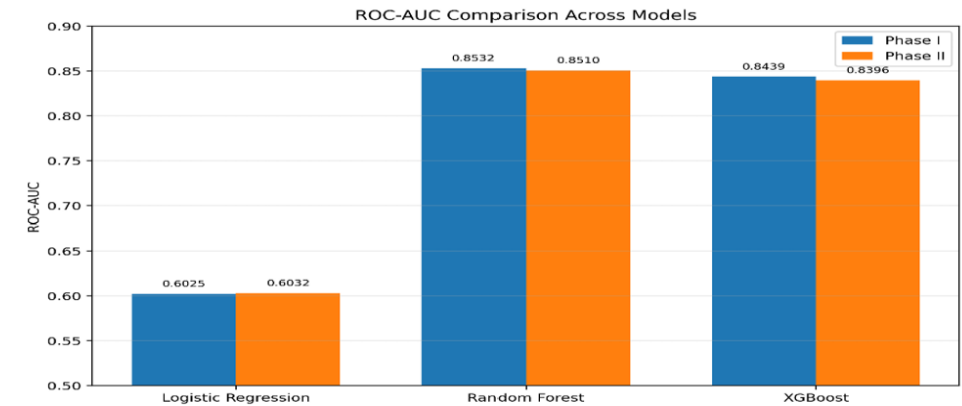
Training Strategy

- ✓ TimeSeriesSplit Cross Validation
- ✓ Hyperparameter tuning
- ✓ Calibration
- ✓ Joblib serialization

The dataset was split chronologically into training and forecast sets to avoid information leakage and mimic real-world forecasting.

Key Finding: Random Forest achieved the best balance between discrimination, calibration, and generalization, while XGBoost achieved the highest accuracy. Overall, Random Forest was selected as the preferred model due to its consistent performance and interpretability.

CV ROC-AUC (higher is better)



ROC-AUC measures discrimination. Random Forest remained the strongest model in both phases.

Comparative Performance of Phase I and Phase II

Model	Metric	Phase I	Phase II	Change
Logistic Regression	ROC-AUC	0.6025	0.6032	Slight Improvement
Logistic Regression	Accuracy	51.67%	73.33%	Improved
Logistic Regression	Recall	44.44%	19.44%	Declined
Random Forest	ROC-AUC	0.8532	0.8510	Slight Decline
Random Forest	Accuracy	87.50%	86.67%	Slight Decline
Random Forest	Recall	75.00%	75.00%	No Change
Random Forest	Brier Score	0.1439	0.1415	Improved
XGBoost	ROC-AUC	0.8439	0.8396	Slight Decline
XGBoost	Accuracy	87.50%	87.50%	No Change
XGBoost	Recall	75.00%	75.00%	No Change
XGBoost	Brier Score	0.1328	0.1325	Improved

Hyperparameter Tuning

- To optimize model performance and improve generalization, hyperparameter tuning was performed using RandomizedSearchCV with TimeSeriesSplit cross-validation.

Examples of tuned parameters include:

- Logistic Regression: Regularization strength (C), penalty, solver
- Random Forest: Number of trees, maximum depth, minimum samples split, minimum samples leaf
- XGBoost: Learning rate, maximum depth, number of estimators, subsample ratio, column sampling ratio

The optimal hyperparameter combinations were selected based on cross-validated ROC-AUC.

Cross-validation and a time-based holdout set were used to assess model generalization and minimize information leakage.

Best hyperparameters (examples)

- LogReg: C=0.1, penalty=L2, solver=saga
- RF: n_estimators=200, max_depth=None, min_samples_split=20, min_samples_leaf=10
- Selected final model: Random Forest
 - ✓ Achieved the highest ROC-AUC, the primary evaluation metric.
 - ✓ Demonstrated the best balance between discrimination, calibration, and generalization.
 - ✓ Showed robust performance under TimeSeriesSplit cross-validation.
 - ✓ Less susceptible to overfitting after hyperparameter tuning.
 - ✓ Supports SHAP-based explainability, making model predictions more interpretable for financial decision-making.
 - ✓ Provides a strong balance between predictive performance and interpretability, making it suitable for deployment as an AI-assisted early warning system.

Final Model Selection

Model	Accuracy	ROC-AUC	Remarks
Logistic Regression	70.8%	0.6009	Baseline model
Random Forest	86.7%	0.8489	Selected Final Model
XGBoost	87.5%	0.8396	Highest accuracy but slightly lower ROC-AUC

Explainability approach

SHAP Level	Brief Explanation
Global Explainability	Identified Interest Expense, Revenue, ROA, Current Assets, and Log(Total Assets) as the most influential predictors.
Local Explainability	Explained why a specific company was predicted as High Risk or Low Risk based on its financial characteristics
Operational Explainability	Supports explainable and trustworthy AI, enabling financial analysts and regulators to justify and communicate model-driven risk assessments.

- SHAP improves model transparency by explaining both overall feature importance and individual predictions, making the AI model more interpretable and suitable for financial decision-making

Limitations**Addressed Limitations**

Class imbalance
Overfitting
Information leakage
Temporal leakage

Mitigation

TimeSeriesSplit
Hyperparameter tuning
Calibration
Generalization Gap
Regularization

Ethical AI & Bias Auditing**Fairness**

Bias audit performed by

- Reviewing feature influence
- Monitoring model calibration
- Comparing prediction consistency across observations

These measures supported a more transparent and reliable AI-assisted decision framework

Model Risk Controls

- Data Leakage Validation – Used a time-based train-test split and TimeSeriesSplit cross-validation to ensure that only historical data were used to predict future outcomes, preventing information leakage.
- Overfitting Prevention – Applied hyperparameter tuning, controlled tree depth, and minimum sample constraints, resulting in a small generalization gap (0.0394).
- Probability Calibration – Evaluated prediction reliability using Brier Score and calibration analysis to ensure predicted probabilities closely reflected actual risk.
- Explainability & Transparency – Used SHAP to explain global feature importance and individual predictions, improving model interpretability and trust.
- Reproducibility & Deployment – Serialized the final model and preprocessing pipeline using Joblib, enabling consistent, reproducible predictions and deployment readiness.

Contributions

- ✓ Developed a multimodal debt risk prediction framework
- ✓ Successfully integrated structured financial data with FinBERT-derived sentiment
- ✓ Demonstrated an explainable AI approach using SHAP

Findings

- ✓ Financial statement variables remained the strongest predictors.
- ✓ Sentiment features provided complementary contextual information rather than replacing traditional financial indicators.
- ✓ Random Forest provided the most balanced overall performance and was selected as the final model for deployment.

Future Work

- ✓ Larger Philippine datasets
- ✓ Alternative LLMs for sentiment extraction
- ✓ Real-time news monitoring
- ✓ Deployment using FastAPI or Streamlit